

Phage therapy for multidrug-resistant *Pseudomonas aeruginosa* infections: a systematic review of the structure and verifiability of the clinical evidence base

Danny Valdiviezo¹, Nataly Trelles¹

¹ School of Medicine, Universidad Católica de Cuenca, Cuenca, Ecuador.

Corresponding author: Danny Valdiviezo. Universidad Católica de Cuenca, Av. de las Américas y Humboldt, Cuenca, Ecuador. Email: dvchiqui[at]gmail.com

Word count: abstract 280; main text 6 374.

Tables: 6. **Figures:** 2. **Supplementary files:** 16 (S0–S15) plus a guide.

Abstract

Background. Several recent syntheses pool phage therapy outcomes for resistant *Pseudomonas aeruginosa* into overall success proportions. That assumes the evidence base can be aggregated, and no one has examined it.

Objectives. To delimit it reproducibly and determine whether its structure and reporting admit a quantitative synthesis of efficacy.

Methods. PRISMA 2020, separate streams for databases and registers. Eight sources — BVS and SciELO included — over a 2016–2026 window where admitted. English or Spanish only, a criterion adopted **after screening closed** and verified against full text. The unit was the study. Extraction was manual, in duplicate and independent, with consensus on disagreements; the structure comes from a pre-extraction from abstracts and the outcomes from the adjudicated extraction.

Results. From 23 057 records, 17 129 unique reports remained; 233 reports formed 184 studies and, on reading the full texts, **29 left the corpus** —protocols, preclinical models, and articles where *P. aeruginosa* appears only in the product's declared spectrum—, leaving **155**, 95 with a retrievable publication. Single case reports, 44.2 %; comparative, 11.6 %; resistance class unassignable, 75.8 %. Full text was obtained for 71 (74.7 %), and what is missing concentrates 54.5 % of the comparative designs; the language criterion removed 35 studies more. In 69 of the 103 arms (67.0 %) no operational definition of clinical success is on record, and only 2 combine comparative design, numerator, denominator and a definition. Setting aside the one whose outcome is a time, **none remains**.

Conclusions. This evidence base is at once large and structurally unsuited to a quantitative synthesis of efficacy: published success proportions rest on assumptions these data do not support.

Keywords: bacteriophages; phage therapy; *Pseudomonas aeruginosa*; multidrug resistance; systematic review; PRISMA.

1. Introduction

Pseudomonas aeruginosa adds to its intrinsic resistance an unusual facility for acquiring new determinants. That combination keeps it among the World Health Organization's priority pathogens, though its position has moved: the 2017 list placed it in the critical tier ¹ and the 2024 update downgraded the carbapenem-resistant form to high priority ². The downgrade does not lessen the clinical problem an isolate with no options poses, but it does mean urgency should not be argued from a superseded classification. Once an isolate accumulates resistance to carbapenems, antipseudomonal cephalosporins, fluoroquinolones and aminoglycosides, the clinician's options can be counted on one hand. Sometimes none is both active and tolerable. The problem concentrates in device-associated infections, extensive burns, cystic fibrosis and bronchiectasis.

Phage therapy has re-emerged into that gap. Lytic bacteriophages act through a mechanism unrelated to antibiotic targets, spare the surrounding microbiota, and can be tailored to a patient's own isolate ³. But the clinical literature supporting them did not grow the way evidence for a drug in development grows. It grew as an accumulation of rescue cases published one at a time, almost always after the patient improved.

That growth pattern carries consequences that adding more cases does not dissolve. A published case report is, by construction, an event conditioned on its outcome. A compassionate-use series recruits patients in whom antibiotics have already failed and to whom the treating team judged it reasonable to offer something

further⁴. And when phage is given alongside active antimicrobials, no favourable result can assign the credit to one or the other⁵.

Published syntheses have nonetheless tended to pool this literature into overall clinical success proportions^{6,7}. That step requires taking three things on trust that are rarely checked. First, that the resistance category used for stratification survives comparison with the criteria it claims to apply. Second, that the pooled outcome names the same thing across studies. Third, that the intervention is homogeneous enough for a common proportion to mean anything.

A prior question therefore precedes any estimate, and it is the one this review addresses: does this evidence base have the structure and the verifiability — the possibility of establishing, from what is published, whether a study is retrievable as a report, what population it concerns and what outcome it measured — that a quantitative synthesis requires? Answering it requires first delimiting the literature completely, including the regional literature and the trial registers that previous reviews have covered unevenly, and then characterising what that body of work states and what it leaves unsaid.

2. Methods

2.1 Protocol and registration

This review was conducted following the PRISMA 2020 statement⁸. The completed checklist accompanies the manuscript as supplementary material. **The review was not registered in PROSPERO or any other prospective registry, and no prior protocol was deposited;** the working protocol and the methodological decisions were date-stamped in the versioned repository from the outset (section 2.8).

2.2 Eligibility criteria

Population. Human patients with *P. aeruginosa* infection classified by the authors as multidrug-resistant (MDR), extensively drug-resistant (XDR) or pandrug-resistant (PDR) according to Magiorakos et al.⁹, or described in terms implying those categories. Multi-pathogen studies were eligible when the *P. aeruginosa* subgroup was separable.

Intervention. Therapeutic administration of lytic bacteriophages by any route, alone or combined with antimicrobials. Endolysins and other phage-derived products given without the viral particle were excluded as a pharmacologically distinct intervention.

Comparator. None required, given the expected nature of the literature.

Outcomes. Clinical success, microbiological eradication, mortality, adverse events and emergence of phage resistance, as each study defined them.

Designs. Any design with treated patients: randomised and non-randomised trials, cohorts, case series and case reports. Excluded were laboratory, preclinical and modelling work; reviews without primary data of their own; secondary syntheses; veterinary infections; and studies of other organisms without a separable subgroup.

Language. Reports written in English or Spanish were eligible. **This criterion was not part of the original protocol and was adopted on 11 August 2026, with screening already closed,** so the decision was taken in knowledge of what the corpus contained. That is precisely what prospective registration exists to prevent, and it is therefore declared as an amendment, with its impact measured, in section 2.9 and among the limitations.

Language was not inferred from the journal name, which fails in both directions: *Pirogov Russian Journal of Surgery* publishes in English, and an English masthead can carry an article in another tongue. It was taken from the language field declared by Europe PMC, the same datum PubMed files under LA, and, where the source did not index it, from the language field of the export itself, which Scopus, SciELO and BVS all carry. Trial registry entries and Cochrane CENTRAL records were treated as eligible on the published language rules of those sources rather than on assumption.

Verification did not stop at metadata. For the studies with a retrieved full text the language was checked against the **whole article**, analysing the central pages rather than the front matter, which many journals

publish in English even when the body is not. That check exposed three reports whose metadata declared English and whose body is in Russian; all three were excluded. The 233 included reports rest on one of three classes of evidence: analysis of their own text, the language rule of the trial register that publishes them, or the source language field. None is unproven. The assignment for each report, with its evidence class and proof, is supplied as supplementary material.

Unit of inclusion. The **study**, not the report, following PRISMA 2020. A protocol, its conference abstracts and its final publication constitute one study with several reports. The decision and its justification are recorded in the project decision log.

2.3 Information sources and search strategy

Eight databases and registers were searched across the two streams PRISMA 2020 requires to be kept separate. As **bibliographic databases**: PubMed, Scopus (in two arms, one organism-first and one intervention-first), Cochrane CENTRAL via Wiley (issue 7 of 12, July 2026), the BVS portal (sixteen collections, among them LILACS, CUMED, BINACIS and MedCarib) and SciELO. As **trial registers**: ClinicalTrials.gov, EudraCT and CTIS. ProQuest was searched and returned 4 416 records, but was not exported and forms no part of this corpus.

The publication window of 2016 to 2026 was applied in PubMed, in both Scopus arms and in CENTRAL; not in BVS, SciELO or the registers, so 495 records earlier than 2016 entered the corpus. Publication-type restrictions applied in PubMed arm A (six document types), in arm B (human studies) and in Scopus arm B (medicine, immunology and pharmacology). No language restriction was applied. **Embase was not searched**, being unavailable under the institutional subscription, nor was the WHO ICTRP. The final run was on 10 August 2026.

The search equations were designed and run by D.V., querying each interface directly; they were neither generated nor executed by the language model whose involvement is declared in section 2.4. They are supplied in full as supplementary material, with the literal syntax of each interface, the number of results and the date, so that anyone can repeat them. BVS and SciELO were included so as not to tilt the review towards the anglophone literature, and contributed 2 of the 233 reports that reached full text: they cover Ibero-America, and of the 48 studies that state their origin, 21 are from Russia, Poland, Georgia or Ukraine.

2.4 Study selection

Deduplication matched DOI, PMID and NCT transitively using union-find. Identifiers were read **only** from identifier fields, never from free text. The reason is concrete: in a trial run that scanned abstracts, four unrelated articles were merged because all four cited the same earlier paper. Records without an identifier were compared by normalised title and year, with no fuzzy matching.

Selection proceeded in three stages. **Stage 1** applied explicit, auditable rules: no title, publication type without primary data, animal scope declared in MeSH, and phage used as a laboratory tool. **Stage 2** screened titles and **stage 3** screened abstracts. Both used a **closed vocabulary of six exclusion codes** (ORG, REV, SEC, LAB, VET, OFF) fixed before screening began. A reason that fits none of the six does not receive an improvised seventh code: the record advances to the next stage instead of being excluded.

Stage 1 is deterministic: a script runs it and every exclusion names the rule that fired. Stages 2 and 3 were conducted by **a single human reviewer (D.V.), without duplication by a second reviewer**. The criteria and the closed vocabulary were fixed before starting; the record-by-record decisions were issued by a large language model (Claude Opus 5, Anthropic) that could only choose among the six codes and could not propose a new one: 13 917 title decisions and 460 abstract decisions. The reviewer examined one by one every report that survived screening before grouping them into studies, and those assignments are supplied in full (S5); the records the model excluded were not re-read. Every decision was logged with its reason and timestamp in an append-only record supplied in full (S3).

Agreement between the model and a human reader was not measured. Against the risk of false negatives the review sets the positive-control audit of section 2.5: after every stage it was verified that none of the 40 studies known in advance lost all of its reports.

Two things the word «extraction» conflates are worth separating. The **pre-extraction from abstracts**, which is the source of this article's structural figures (sections 3.1 to 3.4), was issued by the same model under the variable schema fixed in advance, with every record flagged as partial. The **extraction from full text**, from which section 3.5 and Table 5 derive, was carried out by the two reviewers by hand and without any intervention by the model (section 2.6).

Decision logs are **append-only**. A correction is a new row superseding the previous one, and both remain. This review had to correct one erroneous exclusion, and that correction is auditable only because the original row survived.

2.5 Positive-control audit and screening validation

Before screening began, a set of 40 studies known a priori to be eligible was fixed, drawn from an earlier scoping exercise. After each stage we verified that **no study in that set lost all of its reports**. The check operates at study level rather than report level, because excluding an erratum or a duplicate registry entry belonging to an included study is legitimate, whereas losing the study itself is not. The audit passed at all three stages (40 of 40). During one intermediate run it detected a miscalibrated exclusion rule, which was corrected before proceeding.

Validation over the excluded records. The positive-control audit bounds false negatives only over studies known in advance. To bound them over the unknown ones, a random sample of 350 of the 13 661 records excluded at title or abstract (fixed, reproducible seed) was re-screened in Rayyan, blind to the original verdict. The 6 records marked as includable on the first pass proved, when checked against the criteria of section 2.2, to be laboratory work, modelling work or review without primary data, and were reclassified. **The erroneous-exclusion rate was 0.0 % (0 of 350; exact 95 % CI, 0.00 to 1.05 %)**, which over the frame of 13 661 admits up to 143 lost records. The adjudication is supplied as supplementary material.

2.6 Data extraction

The extraction form, with 23 data variables and controlled vocabularies, was piloted on five studies before use. Extraction was carried out **in duplicate and independently** between the two reviewers. One reviewer extracted the 124 studies that then had a retrievable publication —95 after the exclusions in section 3.1.1— and the second extracted 122 of them, so 98 % of the corpus is double-extracted. Over the 130 comparable arm rows, 541 value disagreements were recorded; agreement, measured before resolving them, was 78 % median agreement and a median Cohen's kappa of 0.62 over the 8 of 9 categorical fields where it is informative, ranging from 64 % to 100 % across variables. Disagreements were resolved by consensus between the two reviewers, with a record of who resolved and when: 525 of 541 are adjudicated. Of the remaining 16, 2 had been closed earlier by a mechanical rule applied to a comparison that was subsequently redone and carry no joint signature, and 14 remain open: 10 because the form offered no "not derivable" option in the fields where the reviewers needed one, and 4 for data requiring the article to be re-read. The complete record, resolution by resolution, is supplied as supplementary material.

The structural characterisation of the corpus (sections 3.1 to 3.4, Tables 1 to 4) derives from a systematic pre-extraction from the abstracts of the 95 current studies, carried out under the same variable schema and flagged as partial in every record. Reporting completeness per outcome (section 3.5, Table 5) derives instead from the adjudicated extraction, and is identified as such for every figure. This report presents no efficacy estimates: it characterises the structure of the evidence base and its **verifiability**, with three components measured — retrievability of the report (sections 3.1 and 3.2), structural reporting completeness (section 3.4) and outcome reporting completeness (section 3.5).

2.7 Risk of bias and certainty

Risk of bias was to be assessed with RoB 2 for randomised trials¹⁰, ROBINS-I for non-randomised studies¹¹ and the Murad domains for case reports and series^{12,13}, and certainty with GRADE¹⁴.

Neither assessment was carried out, and this review does not report them. The reasons are specific and worth stating separately. Risk of bias requires both reviewers to read each article and issue a domain-by-domain judgement; it remains outstanding and is not substituted by any design-derived indicator, which is not the same thing. In addition, 24 of the 95 studies with a retrievable publication still lack full text (section 3.2), so an assessment made today would omit a non-random fraction of the corpus. As for GRADE, it rates the certainty of a pooled estimate, and this review presents none: section 3.5 shows that only 2 of the 103 arms meet the minimum requirements for pooling. Rating the certainty of a result that does not exist would be a formality, not an assessment.

2.8 Synthesis methods and reproducibility

The feasibility of a meta-analysis of proportions was assessed explicitly against the usual requirements of that design^{15,16}: interpretable denominators, an outcome comparable across studies, and a sufficiently homogeneous intervention. The result of that assessment is presented in section 3.6.

The entire pipeline is written as versioned, re-runnable code: search, deduplication, screening, grouping and computation of every figure quoted. Each record is keyed on an identifier derived from its content and never on its position in a list, because a positional key changes referent as soon as the set of sources changes. An automated check reconciles the output of each stage against the input of the next, and fails if any declared source is not represented in the corpus. During development that check uncovered an entire database that had been searched and downloaded but never reached the corpus.

2.9 Protocol amendment: the language criterion

On 11 August 2026, with screening complete and 219 studies already delimited, eligibility was restricted to reports written in English or Spanish. The amendment answers an operational limit of the review team, which cannot extract reliably from a Russian text, and is declared here rather than presented as though it had applied from the outset.

Its effect was measured before it was adopted, and it was not small. It excluded **35 reports** and with them **35 whole studies**, none of which retained a report in an admissible language. The corpus fell from 219 to 184 studies and from 159 to 124 studies with a retrievable publication. The loss was not evenly spread: of 41 comparative designs, 23 remained, and of 21 randomised trials, 16.

Three of those 35 were detected only by reading the article itself. Their metadata declared English; the body is in Russian. The third surfaced only on the publisher's own page, which labels the full text as Russian while the abstract is in English. Had verification stopped at the source language field, all three would have stayed in the review.

The positive-control audit also recorded a concrete casualty. **Ronit et al. (2024)**, a case of phage therapy for a *P. aeruginosa*-infected vascular prosthesis published in Danish in *Ugeskrift for Læger*, belonged to the set of 40 studies fixed a priori as eligible and is excluded by the criterion. This is not a screening failure, since the study was identified and classified correctly, but the price of the criterion, and it is reported because an audit that is only reported when it passes demonstrates nothing.

The figures for the review without the amendment survive intact in the decision log, which is append-only, so any reader can reconstruct the earlier corpus and verify the effect independently.

Second amendment: studies whose full text could not be obtained leave the corpus. Adopted on 2 September 2026, again with screening already closed. It is of a different kind from the first, and that should not be glossed over: every other exclusion reason refers to what the **article says** — no patients, no results, the pathogen is on the product label — whereas this one refers to what **this review could not read**, which is a property of the process, not of the study.

Its measured effect: it excludes **1 study**, a randomised trial of which only a repository copy exists, in a pre-peer-review version and behind an automated-access barrier. Its potential effect is far larger, so declaring it is obligatory: **applied to the 24 studies still without full text, the corpus would fall from 155 to 131 studies, comparative designs from 11 to 5 and randomised trials from 7 to 3**, and the full-text retrieval rate would become 100 % **by construction**. That is, the retrieval bias measured in section 3.2 could no longer be measured, because the criterion eliminates it by definition instead of resolving it. This review does not apply it that way, and the reader should know that the 74.7 % figure coexists with an exclusion that, generalised, would empty it of meaning.

3. Results

3.1 Study selection

The nine sources yielded **23 057 records**, from which 5 928 duplicates were removed to leave **17 129 unique reports** (Figure 1). Stage 1 rules excluded 3 235; of the 13 894 titles screened, 13 434 were excluded and 460 advanced to abstract screening; of those, 227 were excluded and **233 reports proceeded to full text**. Grouped by study, those 233 reports correspond to **184 studies assessed for eligibility**. On reading their full texts during extraction, **29 of them left the corpus** and were excluded; the next subsection details them. The evidence base stands at **158 studies**, grouping 197 reports: 134 with a single report and 24 with several, up to a maximum of eleven reports of one trial.

3.1.1 Twenty-nine studies the screening admitted and that later left

The 29 were excluded for eight reasons, and none is a difference of judgement: they are studies that fail what section 2.2 already required. **Twenty-four** came from reading the article. **Eleven** have *P. aeruginosa* in no patient — it appears in the declared spectrum of the phage product, in the introduction or in the reference list; one of them names no organism at all. **Four** are study protocols: they describe what will be done and report no outcomes. **Three** have no patients: a murine model of ventilator-associated pneumonia, an *in vitro* hollow-fibre model, and a methods paper on preparing phages in the laboratory. **Two** administer something that is not a bacteriophage: an endolysin, and a product claiming to induce the patient's own native phages. **Two** do not evaluate phage therapy in patients: in one the phage is mentioned only as an unavailable option and was never administered, and the other is a reconstructive surgery article where phage therapy appears once, in a table cell. **Two** supply no primary data of their own: an opinion piece with a narrative review, and a two-page editorial by a single author, which the repository indexing it classifies literally as "Editorial" and "not citable".

The **remaining four** were excluded on **language**, and were detected differently. All four appeared in the same Russian journal, and the criterion in section 2.5 had kept them on a premise that turned out to be false: that this journal issues a full English version. It does not. Measured on the publisher's own page and **without machine translation**, the body of all four articles is predominantly Cyrillic — between three quarters and almost all of their characters; what is in Latin script is the abstract, the keywords and the references. The site offers no English version by any of its three routes. One of the four also fails on population, exactly as the ten above do: its patients had *Staphylococcus* and *Streptococcus*, and *P. aeruginosa* appears only in the composition of the pyobacteriophage administered.

That case exposed a defect in the verification procedure itself, and it is worth declaring. The language detector analyses the title and the **indexed abstract**, yet its strongest evidence class was named "text proven". For a study whose full text was never retrieved that asserts more than was checked, and it applies to **29 of the 95 extractable studies**. When the source declares a non-admitted language and the abstract is in English, what is proven is that the journal publishes an English abstract — the norm across much of Eastern

Europe — not that the article can be read in English. The class was renamed and that precedence corrected.

The cause is common to nearly all of them and is worth stating plainly: title and abstract screening was issued by a language model as sole reviewer (section 2.4). A protocol abstract, a murine-model abstract and one listing *P. aeruginosa* in the spectrum of a polyvalent phage **look very much** like an eligible abstract; telling them apart requires opening the article. Section 2.4 declares that the records the model excluded were never re-read by a human. These 22 are the other side of that limitation, and they are **23.7 %** of the 93 studies whose full text could be read. It is a floor, not a total: the 31 studies without full text could not be checked against anything.

Of the 155 studies, **95 have a retrievable publication**: 92 articles and 3 that exist only as conference abstracts. The remaining **60 are trial registry entries alone**, with no locatable associated publication. That figure by itself describes a field in which more than one study in three has been registered but has never published results.

Exclusion reasons at title screening were laboratory or preclinical work (4 060), other organism without a separable subgroup (3 022), review or commentary without primary data (2 803), does not evaluate phage therapy in patients (2 631), veterinary scope (834) and secondary synthesis (84). At abstract screening: other organism (65), review or commentary (64), does not evaluate phage therapy (40), **inadmissible language (35)**, laboratory (18) and secondary synthesis (5).

3.2 Full-text retrieval and its bias

Full text was obtained for **71 of the 95 retrievable studies (74.7 %)** through legitimate open-access routes. The remaining 24 have not been obtained. Their availability was re-checked on 2 September 2026 against three independent indexes — Europe PMC, OpenAlex and Semantic Scholar — and **none of the 24 has an open-access copy today**. They were not requested through interlibrary loan or from the authors: the documentary gap this review declares is not mitigated by any further route, which is why it is reported as a limit and not as something pending.

The unretrieved fraction **is not a random sample of the whole** (Table 3). It contains 6 of the 11 comparative studies, **54.5 %**, and, although it reports fewer patients overall, it concentrates the designs that can sustain a comparison. Of the field's four landmark randomised trials^{17,18,19,20}, two remain unread: the intravesical phage trial and TP-102. The PhagoBurn article was obtained. The Yale trial has no full journal article, only a conference abstract and results deposited in the register; what was obtained is the trial protocol, which documents the design but not the outcomes.

The asymmetry is itself a finding. Full-text accessibility covaries with region of publication and with study design, and it pushes in the same direction as the language restriction: both tilt the corpus towards the anglophone open-access literature. A review that declares neither transforms the question without saying so.

3.3 Characteristics of the evidence base

The 95 retrievable studies were published between **2016 and 2026**, and **80.0 %** since 2020. This is a young and rapidly expanding evidence base (Table 1).

Its composition by design is what a rescue literature produces, not what a synthesis of efficacy needs: **42 single case reports (44.2 %)**, 18 case series, 13 prospective cohorts, 7 randomised trials, 4 non-randomised trials and 3 retrospective cohorts. A further 8 do not state a recognisable design in the abstract. Comparative designs total **11 studies, 11.6 %** of the whole, and are the only ones able to sustain a claim of relative efficacy.

Where stated, geographic origin is spread across Russia (7 studies), Poland and Georgia (6 each), Germany (4), France and the United States (3 each), and India, Iran and Israel (2 each). It is not stated in 76 studies. The fall in Russian representation relative to the pre-amendment corpus, from 34 studies to 7, is a direct effect of the language criterion and not of the search.

3.4 What the evidence base does not state

Reporting completeness, measured on what abstracts make explicit, is lowest precisely in the variables on which any stratification depends (Table 2):

- **Resistance class** (MDR, XDR or PDR) cannot be assigned in **75.8 %** of studies: 62.1 % do not mention it at all and a further 13.7 % mention it in terms that do not allow classification. For stratifying by resistance, which is what published syntheses do, both cases are equally unusable.
- **Route of administration** is not stated in **52.6 %**.
- **Pathogen scope**, that is, whether the study covers only *P. aeruginosa* or is mixed with a separable subgroup, is not stated in **38.9 %**.
- Fulfilment of the **DTR** criterion (difficult-to-treat resistance) is not stated in **97.9 %**.

These figures describe reporting in the abstract, not necessarily in the full article, and the distinction matters because some of the information may sit in the body of the text. That said, the abstract is what databases index and what automated reviews consume. Three studies in four failing to state their resistance category there means that no MDR/XDR/PDR stratification built from abstracts can be correct.

3.5 What the corpus reports for each outcome

Duplicate extraction, now adjudicated, makes it possible for the first time to measure what this evidence base reports for the five outcomes declared in the eligibility criteria. Table 5 summarises it over the 132 extracted arms.

Completeness is uneven and, at best, partial. Adverse events are the outcome most often reported with a denominator (77.7 % of arms) and emergence of phage resistance the least (27.2 %), despite being the outcome that determines whether the intervention retains its usefulness. Those percentages are computed over all 103 arms and therefore include 24 from studies whose full text has not been retrieved: of an article one has not been able to read, one cannot assert that it is silent. Over the 79 readable arms the figures rise — from 74.5 % to 88.6 % for adverse events, from 25.5 % to 35.4 % for resistance — and Table 5 gives both columns so the reader can separate the literature's silence from this review's own documentary gap. Clinical success, the outcome that published syntheses pool, is reported with numerator and denominator in 68.0 % of arms. Two arms report a numerator larger than their denominator — thirteen clinical successes in one patient, and five adverse events in four — and are flagged as reporting errors, not read as proportions.

That 68.0 % is misleading, however, if read as availability for pooling. **In 69 of the 103 arms (67.0 %) no operational definition of clinical success is on record**; over the 79 readable arms, in 45 (57.0 %). In 48 of them the reviewers established this by reading the article: 47 state a number of patients with a "good outcome" without saying what had to be observed to count one, and 1 left the field empty. In the remaining 21 the extraction is incomplete, so neither the presence nor the absence of a definition is on record; they are counted among those that do not supply one because the datum is not available to a reader of this review, not because its absence was established. A numerator without a definition names nothing comparable to another study's numerator.

Study design is not always on record either: in 21 of the 103 arms it cannot be classified — 13 where both reviewers agreed on "NA" and 8 still open — so the design counts are floors over the remaining 82. Of those

82, **7 arms are trials** (randomised or not) and **9 are cohorts**. Section 3.3 and Table 1 call only the trials comparative, which is why the figure there is 23 studies; when this section says comparative it also includes the cohorts, and there are 16 arms. They are not the same set.

Crossing the requirements a pooled proportion would demand empties the corpus step by step. Four of them are arithmetic — comparative design, numerator, denominator and an operational definition — and leave **2 of the 103 arms**. The next two are the eligibility criteria this review itself declares in section 2.2, and none of those 3 survives them.

The first is that the outcome be attributable to *P. aeruginosa*. Section 2.2 admits multi-pathogen studies only when the subgroup is separable, and in one of the three the extraction records that the article does not separate the data: its numerator counts patients of several species. Two remain. The second is that administration be therapeutic. The next of the three is a trial *preventing* ventilator-associated pneumonia, and its outcome — absence of incident pneumonia — does not measure the same thing as cure of an established infection. One remains.

That last one is PhagoBurn, whose primary outcome is the *time* to a sustained reduction in bacterial burden¹⁷: not a proportion, and one that would enter a meta-analysis of proportions only if the synthesist converted it into one. **None remains.**

Put another way: of the 103 arms of this literature, **not one meets both the arithmetic conditions and the eligibility criteria that would make its clinical success a legitimate term in a pooled proportion**. It is not that the available studies are few: there are none. Table 6 sets out the funnel, step by step.

3.6 Feasibility of quantitative synthesis

A meta-analysis of proportions requires interpretable denominators, an outcome that names the same thing across studies, and a sufficiently homogeneous intervention. The body of evidence delimited here fails all three requirements.

Denominators. Almost four studies in ten are single cases, in which the success proportion can only take the values 0 or 1 and its binomial variance is maximal. Pooling forty-nine denominators of one with a handful of larger series produces a weighted mean whose weights are set by series size rather than by evidential quality.

Outcome. Among the studies whose full text has been read, "clinical success" neither names the same thing nor measures on the same scale. At one end, composite long-term clinical cure: no implant loosening and walking without pain at 21 months²¹. At the other, discharge criteria: patient discharged, de-cannulated and rehabilitated⁴. In between, the label without a criterion: "good clinical outcome" declared by the authors as binary, without stating what had to be observed²².

The problem is not only heterogeneity. **Two of the landmark randomised trials report no binary success proportion at all**: the primary endpoint of PhagoBurn is the *time* to a sustained reduction in bacterial burden¹⁷, and that of BX004-A is a *continuous* variable, the mean log-scale CFU reduction at day 15²³. A meta-analysis of proportions cannot include them without binarising what their authors measured otherwise, and that conversion would be made by the synthesiser, not by the study.

These definitions come from the full texts already retrieved, not from the pre-extraction from abstracts: **no abstract in the corpus states a definition of clinical success**, which is itself a datum about how the field reports.

Intervention. The products range from fixed-composition commercial polyvalent preparations to cocktails tailored to a patient's own isolate, and in most cases they are co-administered with active antimicrobials. The heterogeneity is conceptual rather than statistical, and no random-effects model resolves it: averaging different interventions yields a number that corresponds to no real intervention.

Added to this is the retrieval bias documented in section 3.2, which as things stand would exclude two-thirds of the comparative studies from any synthesis. **For these reasons no pooled efficacy proportion is presented here**, and we argue that those published in previous reviews rest on assumptions these data do not

verify.

4. Discussion

4.1 Principal findings

This review delimits the clinical evidence base for phage therapy in resistant *P. aeruginosa* more broadly than previous syntheses: nine sources, no language restriction, regional literature and trial registers included. The result is 155 studies against the dozens handled by published reviews, and would have been 219 without the language restriction adopted at the end and without the 29 that later left. That increase does not speak well of the field's maturity. Most of it comes from case reports, from registry entries without publication, and from regional literature that searches confined to PubMed and Scopus do not reach.

The central result is negative and worth stating plainly: **the breadth of this evidence base does not translate into capacity for synthesis**. Almost four retrievable studies in ten are a single case. Fewer than one in five is comparative. In four out of five, the resistance category by which stratification is attempted cannot be assigned. And almost one identified study in three exists only as a registry entry, with no published results.

4.2 Comparison with previous literature

Recent reviews have reported overall clinical success proportions consistently above 80 %^{6,7}. Those results do not contradict ours; they explain them. A literature composed largely of cases published after a favourable outcome yields, when pooled, a high proportion by construction. What that proportion measures is the propensity to publish successes, not the efficacy of the intervention.

Our delimitation further suggests that those syntheses worked on a fraction of the field. The unrestricted search retrieved 34 Russian and 3 Ukrainian studies, and that is precisely where comparative designs concentrate. The language amendment obliges us to set them aside, so this review documents that body of literature and where it sits but cannot speak to its content. That is an explicit debt to the next review with translation capacity.

4.3 Implications

For clinical practice. No published success proportion for phage therapy in resistant *P. aeruginosa* should be used to estimate the probability of benefit for an individual patient. The available evidence supports the statements that phage therapy is being given, that it is tolerated, and that favourable outcomes have been described. It does not support an effect size.

For research. Three shortcomings could be remedied at no added cost and would transform the usefulness of the next reports. State the resistance category together with the explicit criteria used to assign it. Document that the administered phage had verified lytic activity against the patient's isolate. And define clinical success with an explicit time horizon. The first is the most urgent: without it, the population of the question is undefined.

For registries. That 60 of 155 studies are registry entries without a locatable publication is a first-order publication-bias problem, and no statistical method resolves it.

4.4 Strengths and limitations

Strengths. The search covered nine sources across two separate streams, with no language restriction. Selection used a closed vocabulary fixed in advance and was audited against a positive-control set defined before screening began, which allows us to state, rather than merely hope, that screening lost no known eligible study. The whole pipeline is re-runnable code with content-derived keys, and an automated check reconciles each link with the next.

Limitations. There are six, and all are material. **First**, the corpus excludes one study for the sole reason that its full text could not be read (section 2.9); applied consistently to the 24 others without text, that criterion would empty the measurement of retrieval bias of content. It is declared here because an exclusion that depends on the reviewer's access, rather than on the study's content, is not comparable with the rest. **Second**, the search did not query Embase or the WHO ICTRP, which reduces coverage of case series and conference abstracts and leaves the register stream limited to the United States and the European Union. **Third**, it was limited to

publications from 2016 to 2026, so it describes recent output rather than the whole field. **Fourth**, this report describes how completely outcomes are reported (section 3.5) but presents no efficacy estimates, no risk-of-bias assessment and no GRADE certainty (section 2.7): it is a characterisation of the evidence base, not a synthesis of its results. **Fifth**, full text was obtained for 74.7 % of the retrievable studies, and the missing fraction is enriched in comparative designs, so what is described under-represents the comparative part of the field. **Sixth**, the restriction to English and Spanish was adopted once screening had closed and removed 35 studies, 18 of them comparative and one from the positive-control set; it pushes the corpus in the same direction as the retrieval bias and narrows what this review can say about the literature of Eastern Europe.

5. Conclusions

The clinical evidence base for phage therapy in resistant *P. aeruginosa* is larger than published syntheses reflect and, at the same time, less suited to quantitative synthesis than those syntheses assume. It consists mainly of single cases and non-comparative series, published conditionally on their outcome, with a reporting completeness that does not allow verification of the resistance category defining the population of interest. While those three conditions hold, pooling their outcomes into a common proportion yields a figure that admits no clinical interpretation.

Declarations

Funding. This review received no specific funding from public, commercial or not-for-profit agencies.

Ethics approval and consent. Not applicable. This review synthesises already-published literature and aggregate data from public trial registers; it involves no patients, no specimens and no identifiable individual data, and therefore required neither ethics committee approval nor informed consent. No formal waiver was sought because the institution does not require one for studies without primary data.

Competing interests. The authors declare no competing interests.

Author contributions (CRediT). **D. Valdiviezo:** conceptualisation, methodology, investigation (design and execution of the searches across the nine sources), software (screening, grouping and figure-computation pipeline), data curation, data extraction, writing — original draft, visualisation, project administration. **N. Trelles:** methodology (outcome definitions), investigation, independent data extraction (122 of the 124 studies), validation, writing — review and editing.

Data and code availability. All material supporting this review is deposited in a public, citable repository: the search corpus with the raw exports of all nine sources, the append-only decision logs of the three screening stages, both reviewers' extraction workbooks, the methodological decision records with their dates, and the complete pipeline code, which is re-executable from search to the figures in the manuscript. «DEPOSIT DOI — TO BE COMPLETED BEFORE SUBMISSION». The decision logs are append-only and preserve every correction alongside the row it supersedes.

Use of artificial intelligence. Screening was conducted by a single human reviewer, without duplication by a second reviewer. A generative language model (Claude Opus 5, Anthropic) was used to issue the record-by-record decisions of title and abstract screening and for the pre-extraction from abstracts, applying criteria and a closed vocabulary fixed in advance by the authors, and in programming and drafting assistance. Reports that survived screening were reviewed one by one by the authors; those the model excluded were not re-read by a human. The design and execution of the searches, the fixing of the eligibility criteria and the data extraction were carried out without the model. Every screening decision is logged with its reason and timestamp and is supplied in full (S3, S4). The authors take responsibility for the accuracy and integrity of the work. No AI tool is listed as an author, in line with ICMJE recommendations.

Tables and figures

The six tables are supplied as separate files (paper/tables/), each with its own footnote. Both figures are supplied as vector PDF (paper/figuras/).

Table 1. Characteristics of the retrievable evidence base (n = 95 studies).

Table 2. Reporting completeness for variables critical to stratification.

Table 3. Comparison of studies with and without full text obtained.

Table 4. Exclusion reasons by stage, using the closed vocabulary.

Table 5. Reporting completeness for the five declared outcomes, over the adjudicated extraction (n = 103 arms).

Table 6. Arms surviving each requirement for a pooled proportion of clinical success (n = 103 arms).

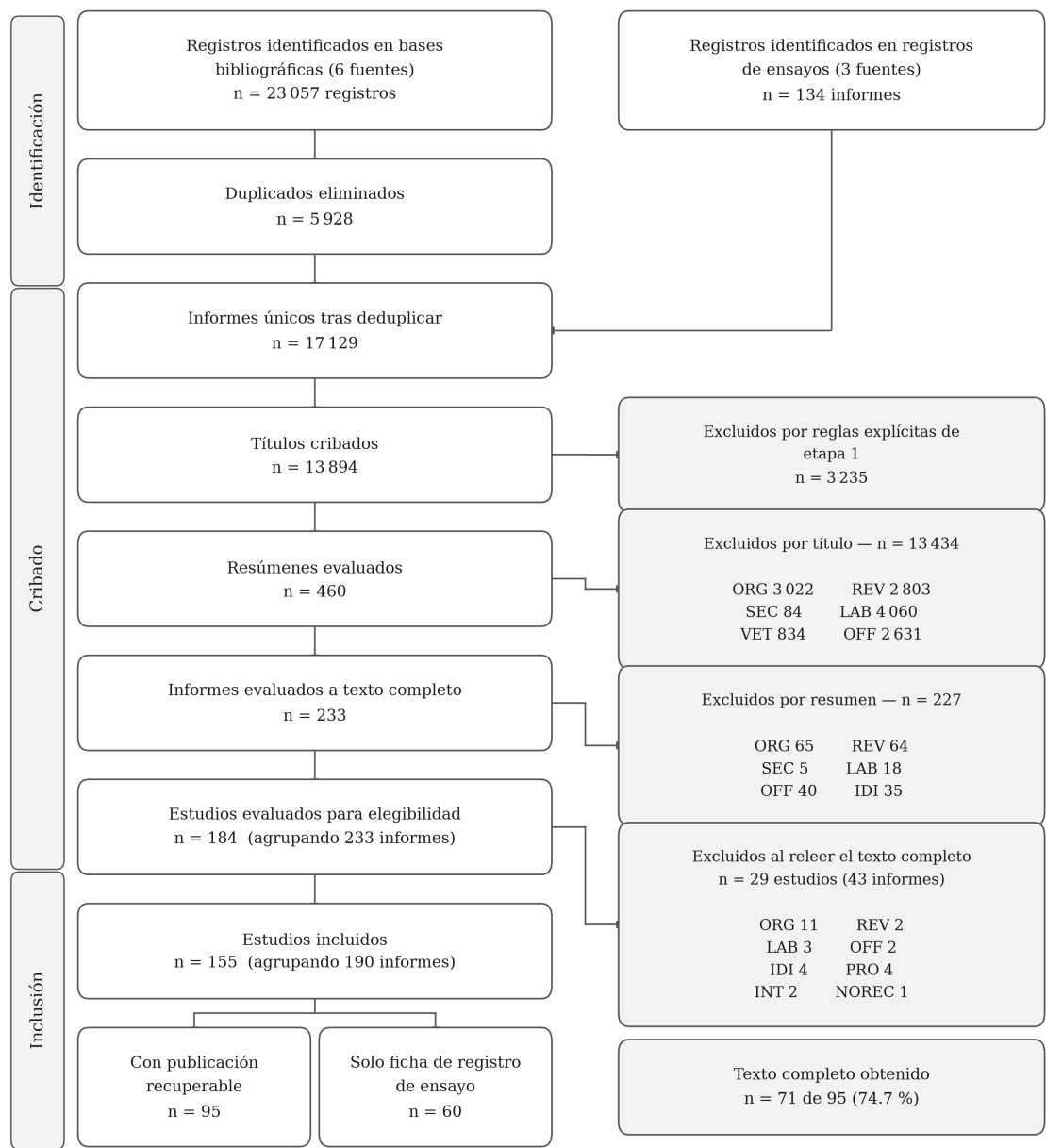


Figure 1. PRISMA 2020 flow diagram.

What it shows. The passage from identified records to included studies, with the two identification streams — bibliographic databases and trial registers — counted separately, as the PRISMA 2020 statement requires. *How to read it.* The left column is the main flow; the grey boxes on the right hold what was excluded at each step, broken down by the closed six-code vocabulary (ORG, REV, SEC, LAB, VET, OFF; defined in Table 4). The register stream enters from the side and does not pass through bibliographic deduplication, because a registry entry has no DOI with which to collide against an article. The final grey box is not part of the selection flow: it reports how many included studies have been read in full text to date. *Source.* Project screening pipeline; final search run, 10 August 2026.

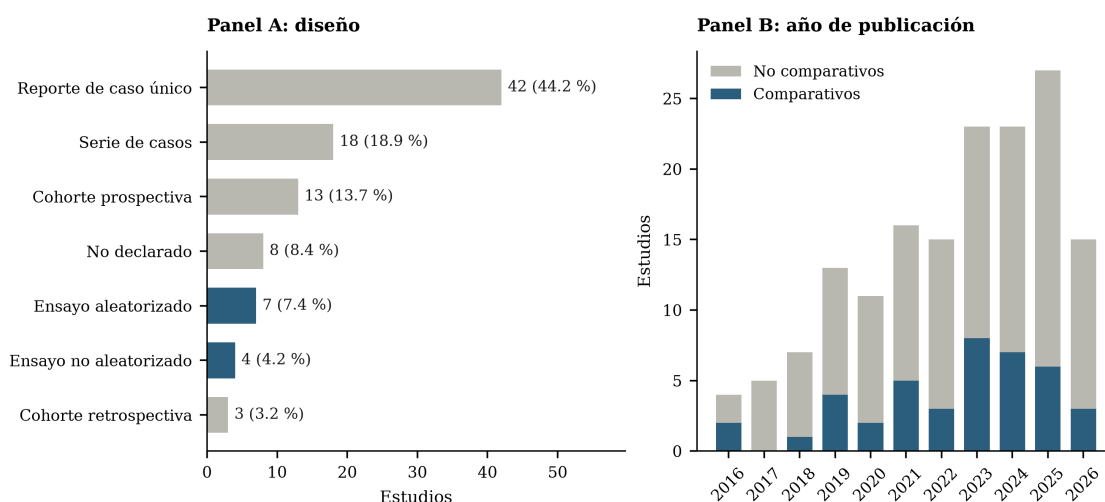


Figure 2. Composition of the retrievable evidence base (n = 95 studies).

What it shows. Panel A orders studies by design; Panel B distributes them by year of publication, separating comparative designs from the rest within each year. *How to read it.* In both panels the dark colour marks comparative designs — randomised and non-randomised trials — the only ones able to sustain a claim of relative efficacy. "Not stated" in Panel A means the abstract does not allow the design to be recognised, not that the study lacks one. The year 2026 is incomplete: the search closed in August. *Source.* Systematic pre-extraction from the abstracts of the 95 studies with a retrievable publication.

Supplementary material

- **S0.** Informe editorial: qué se decidió, cuándo y con qué fundamento.
- **S1.** Lista de comprobación PRISMA 2020.
- **S2.** Ecuaciones de búsqueda literales por fuente, con fecha y número de resultados.
- **S3.** Registro completo de decisiones de cribado, en dos ficheros: etapa 2 (título) y etapa 3 (resumen). Cada fila lleva su motivo codificado, quién la emitió y su marca de tiempo.
- **S4.** Pre-extracción desde el resumen, marcada como parcial en cada registro.
- **S5.** Listado de los 184 estudios con su situación y sus informes agrupados.
- **S6.** Auditoría de controles positivos.
- **S7.** Declaraciones ICMJE.
- **S8.** Registro de recuperación de texto completo: qué se intentó por cada estudio y con qué resultado.
- **S9.** Idioma por informe, con la clase de evidencia que aportaba.
- **S10.** Verificación del idioma sobre el texto completo del PDF, no sobre los metadatos.
- **S11.** Concordancia entre las dos extracciones independientes, antes de resolver los desacuerdos.
- **S12.** Resolución de conflictos de extracción, incluidos los que no pueden dirimirse todavía.
- **S14.** Adjudicated extraction: one value per cell, arm by arm, after consensus. Underpins section 3.5 and Tables 5 and 6.
- **S15.** Provenance of every cell in S14: agreement, consensus, single reading, or still open.
- **S13.** Reglas de extracción de desenlaces, con su corrección documentada.

Los acompaña una guía que explica, fichero a fichero, qué contiene y qué pregunta permite contestar.

References

1. Tacconelli E, Carrara E, Savoldi A, Harbarth S, Mendelson M, Monnet DL, et al. Discovery, Research, and Development of New Antibiotics: The WHO Priority List of Antibiotic-Resistant Bacteria and Tuberculosis. *Lancet Infectious Diseases*. 2018;18(3):318-327. doi:10.1016/S1473-3099(17)30753-3
2. World Health Organization. WHO Bacterial Priority Pathogens List, 2024: Bacterial Pathogens of Public Health Importance to Guide Research, Development and Strategies to Prevent and Control Antimicrobial Resistance. 2024.
3. Pirnay J, Djebara S, Steurs G, Griselain J, Cochez C, De Soir S, et al. Personalized Bacteriophage Therapy Outcomes for 100 Consecutive Cases: A Multicentre, Multinational, Retrospective Observational Study. *Nature Microbiology*. 2024;9(6):1434-1453. doi:10.1038/s41564-024-01705-x
4. Aslam S, Courtwright AM, Koval C, Lehman SM, Morales S, Furr C, et al. Early Clinical Experience of Bacteriophage Therapy in 3 Lung Transplant Recipients. *American Journal of Transplantation*. 2019;19(9):2631-2639. doi:10.1111/ajt.15503
5. Chan BK, Turner PE, Kim S, Mojibian HR, Elefteriades JA, Narayan D. Phage Treatment of an Aortic Graft Infected With *Pseudomonas aeruginosa*. *Evolution, Medicine, and Public Health*. 2018;2018(1):60-66. doi:10.1093/emph/eoy005
6. Uytendaele S, Chen B, Onsea J, Ruythooren F, Debaveye Y, Devolder D, et al. Safety and Efficacy of Phage Therapy in Difficult-to-Treat Infections: A Systematic Review. *Lancet Infectious Diseases*. 2022;22(8):e208-e220. doi:10.1016/S1473-3099(21)00612-5
7. Liu Y, Thong S, Moreira W, Yeo JH, Zhong Y, Chong ZS, et al. Clinical Application of Customized and Non-Customized Bacteriophage Therapy in Patients with Refractory/Resistant Bacterial Infections: A Systematic Review and Meta-Analysis. *International Journal of Antimicrobial Agents*. 2025;66(4):107570. doi:10.1016/j.ijantimicag.2025.107570
8. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71
9. Magiorakos A, Srinivasan A, Carey RB, Carmeli Y, Falagas ME, Giske CG, et al. Multidrug-Resistant, Extensively Drug-Resistant and Pandrug-Resistant Bacteria: An International Expert Proposal for Interim Standard Definitions for Acquired Resistance. *Clinical Microbiology and Infection*. 2012;18(3):268-281. doi:10.1111/j.1469-0691.2011.03570.x
10. Sterne JAC, Savovi\c J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: A Revised Tool for Assessing Risk of Bias in Randomised Trials. *BMJ*. 2019;366:l4898. doi:10.1136/bmj.l4898
11. Sterne JA, Hernan MA, Reeves BC, Savovi\c J, Berkman ND, Viswanathan M, et al. ROBINS-I: A Tool for Assessing Risk of Bias in Non-Randomised Studies of Interventions. *BMJ*. 2016;355:i4919. doi:10.1136/bmj.i4919
12. Murad MH, Sultan S, Haffar S, Bazerbachi F. Methodological Quality and Synthesis of Case Series and Case Reports. *BMJ Evidence-Based Medicine*. 2018;23(2):60-63. doi:10.1136/bmjebm-2017-110853
13. Munn Z, Barker TH, Moola S, Tufanaru C, Stern C, McArthur A, et al. Methodological Quality of Case Series Studies: An Introduction to the JBI Critical Appraisal Tool. *JBIC Evidence Synthesis*. 2020;18(10):2127-2133. doi:10.11124/JBISIR-D-19-00099
14. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE Guidelines: 1. Introduction --- GRADE Evidence Profiles and Summary of Findings Tables. *Journal of Clinical Epidemiology*. 2011;64(4):383-394. doi:10.1016/j.jclinepi.2010.04.026
15. Barker TH, Migliavaca CB, Stein C, Colpani V, Falavigna M, Aromataris E, et al. Conducting Proportional Meta-Analysis in Different Types of Systematic Reviews: A Guide for Synthesizers of Evidence. *BMC Medical Research Methodology*. 2021;21(1):189. doi:10.1186/s12874-021-01381-z
16. Migliavaca CB, Stein C, Colpani V, Barker TH, Ziegelmann PK, Munn Z, et al. Meta-Analysis of Prevalence: I^2 Statistic and How to Deal With Heterogeneity. *Research Synthesis Methods*. 2022;13(3):363-367. doi:10.1002/jrsm.1547
17. Jault P, Leclerc T, Jennes S, Pirnay J, Que Y, Resch G, et al. Efficacy and Tolerability of a Cocktail of Bacteriophages to Treat Burn Wounds Infected by *Pseudomonas aeruginosa* (PhagoBurn): A Randomised, Controlled, Double-Blind Phase 1/2 Trial. *Lancet Infectious Diseases*. 2019;19(1):35-45. doi:10.1016/S1473-3099(18)30482-1
18. Leitner L, Ujmajuridze A, Chanishvili N, Goderdzishvili M, Chkonia I, Rigvava S, et al. Intravesical Bacteriophages for Treating Urinary Tract Infections in Patients Undergoing Transurethral Resection of the Prostate: A Randomised, Placebo-Controlled, Double-Blind Clinical Trial. *Lancet Infectious Diseases*. 2021;21(3):427-436. doi:10.1016/S1473-3099(20)30330-3

19. Nir-Paz R, Onallah H, Dekel M, Gellman Y, Haze A, Ben-Ami R, et al. Randomized double-blind study on safety and tolerability of TP-102 phage cocktail in patients with infected and non-infected diabetic foot ulcers. *Med.* 2025;6:100565. doi:10.1016/j.medj.2024.11.018
20. Chan B, Kortright K, Stanley GL, Cochrane C, Lee A, Vill A, et al. CYstic Fibrosis bacterioPHage study at Yale (CYPHY). *Journal of Cystic Fibrosis.* 2023;22:S22. doi:10.1016/S1569-1993(23)00976-1
21. Ferry T, Kolenda C, Laurent F, Leboucher G, Merabishvili M, Djebara S, et al. Personalized Bacteriophage Therapy to Treat Pandrug-Resistant Spinal *Pseudomonas aeruginosa* Infection. *Nature Communications.* 2022;13:4239. doi:10.1038/s41467-022-31837-9
22. Onallah H, Hazan R, Nir-Paz R, others. Refractory *Pseudomonas aeruginosa* Infections Treated with Phage PASA16: A Compassionate Use Case Series. *Med.* 2023;4(9):600-611.e4. doi:10.1016/j.medj.2023.07.002
23. Weiner I, Kahan-Hanum M, Buchstab N, Zelcbuch L, Navok S, Sherman I, et al. Phage Therapy with Nebulized Cocktail BX004-A for Chronic *Pseudomonas aeruginosa* Infections in Cystic Fibrosis: A Randomized First-in-Human Trial. *Nature Communications.* 2025;16(1):5579. doi:10.1038/s41467-025-60598-4